ELECTRODYNAMICS II · SEMESTER 5

Electromagnetic waves

Module 3 · PYQ + model paper

40 questions · 20 short + 16 problems + 4 essays
Double the question count of the supplied April 2026 FYUGP prerequisite paper.

CU-FYUGP syllabus 2024 · study edition
SI units · phone pages · October 2026

Practice instructions

Suggested time: two sittings of two hours. Section A: 20 questions × 3 marks, ceiling 48. Section B: 16 × 6, ceiling 72. Section C: answer any two of four × 10. Maximum: 140.

This doubles the supplied April 2026 **Electrodynamics I** 70-mark practice pattern. The current ED II syllabus specifies 70 theory marks but does not supply a question pattern here. This is a study paper, not an official FYUGP mock.

Each numbered question has a distinct primary assessment target. Longer questions combine earlier methods. N25 means November 2025 CBCSS ED I; A26 means April 2026 FYUGP ED I. Tags explicitly identify these older or prerequisite courses. Expanded/adapted means wording, scope or mark allocation was changed. Overlapping past questions are tagged with their original course and number; untagged questions are original model practice.

All four theory modules are included in the set. Module V is laboratory work and is outside this theory-paper series.

Section A · Short answers

3 marks each · ceiling 48

1. Determine the direction and speed of $f(z,t) = A\cos(4z + 12t)$, with SI coordinates.

Model · syllabus-based

2. Write the one-dimensional wave equation and interpret its two derivatives.

Model · syllabus-based

3. A vacuum wave has f=100 MHz. Find λ, k and ω.

Model · syllabus-based

4. Two points along a sinusoidal wave are separated by λ/4. What is their phase difference at the same time?

Model · syllabus-based

5. Define polarization and explain why a vacuum plane EM wave can be polarized.

Model · syllabus-based

6. What amplitude and phase conditions produce circular polarization from two perpendicular components?

Model · syllabus-based

7. Classify $E_x = 3\cos\phi$ and $E_y = 4\cos\phi$. Give peak magnitude and orientation.

Model · syllabus-based

8. Classify $E_x = 3\cos\phi$, $E_y = 4\sin\phi$ and write the tip's locus.

Model · syllabus-based

9. How does Maxwell theory predict the speed of waves in vacuum?

Model · syllabus-based

10. State the assumptions needed for the elementary vacuum EM wave equation.

Model · syllabus-based

11. A wave travels along +x and E points +z at an instant. Which way does B point?

Model · syllabus-based

12. A vacuum plane wave has electric peak 300 V/m. Find its magnetic peak and relative phase.

Model · syllabus-based

13. What does transverse mean for a vacuum plane electromagnetic wave?

Model · syllabus-based

14. Define the vacuum Poynting vector and state its physical meaning and SI unit.

Model · syllabus-based

15. Interpret positive $\mathbf{J} \cdot \mathbf{E}$ in Poynting's theorem.

Model · syllabus-based

16. In a vacuum travelling plane wave, compare electric and magnetic energy densities.

Model · syllabus-based

17. Express the average intensity of a vacuum sinusoidal wave using peak and rms electric fields.

Model · syllabus-based

18. What is the pressure on an ideal absorbing surface at normal incidence from a vacuum beam of intensity I?

Model · syllabus-based

19. Why does an ideal reflector have twice the normal radiation pressure of an ideal absorber at the same incident intensity?

Model · syllabus-based

20. A lossless nonmagnetic dielectric has $\epsilon_r = 2.25$. Find refractive index and wave speed.

Model · syllabus-based

Section B · Problems and explanations

6 marks each · ceiling 72

21. Show directly that any twice-differentiable $f(z - vt)$ solves the one-dimensional wave equation.

Model · syllabus-based

22. Write E for a +z travelling, x-polarized vacuum wave with peak 20 V/m and wavelength 0.60 m, whose field is maximum at z=t=0. Find B.

Model · syllabus-based

23. For $E_x = a\cos\phi$ and $E_y = b\cos(\phi + \delta)$, derive the polarization ellipse equation.

Model · syllabus-based

24. At z=0 take $E_x = E_0 \cos \omega t$, $E_y = E_0 \sin \omega t$. Describe its rotation as viewed from +z looking toward the origin. Avoid an unqualified handedness label.

Model · syllabus-based

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25. Derive the source-free vacuum wave equation for E from Maxwell's equations.

Model · syllabus-based

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26. Derive the source-free vacuum wave equation for B, stating the identity and divergence law used.

Model · syllabus-based

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27. For $E = E_0 \cos(kz - \omega t)\hat{x}$, use Faraday's law to derive B for a vacuum travelling wave.

Model · syllabus-based

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28. Someone proposes a +z vacuum wave with E along +x and B along +y but $B_0 = 2E_0/c$. Can the pair satisfy Maxwell's equations? Explain.

Model · syllabus-based

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29. A region loses field energy at 5 W while charges inside receive 2 W from the field. Find the net outward electromagnetic power.

Model · syllabus-based

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30. A vacuum plane wave has $E_0 = 100$ V/m. Calculate mean energy density and intensity.

Model · syllabus-based

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31. A uniform vacuum beam carries 2 W through area 1.0×10^{-4} m². Find intensity and peak E and B.

Model · syllabus-based

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32. A 3 W beam hits a surface normally in vacuum. Find force for complete absorption and perfect reflection.

Model · syllabus-based

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33. At normal incidence a surface reflects fraction R of a vacuum beam's intensity and absorbs the remainder, with no transmission. Derive its pressure.

Model · syllabus-based

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34. A lossless homogeneous medium has $\epsilon_r = 4$, $\mu_r = 1$ and electric peak 60 V/m. Find speed, impedance, B_o, H_o and intensity.

Model · syllabus-based

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35. A 600 nm vacuum wave enters a stationary transparent nonmagnetic medium with n=1.5. Find f, v and wavelength inside. Explain which quantity stays fixed.

Model · syllabus-based

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36. Show that S=vu for a travelling plane wave in a homogeneous linear lossless nondispersive medium.

Model · syllabus-based

Section C · Essays

10 marks each · answer any two

37. Starting with source-free Maxwell equations, derive the wave speed, transversality and electric/magnetic amplitude relation of a monochromatic plane wave.

Model · syllabus-based

38. Derive Poynting's theorem in vacuum with charges and currents, then give its integral interpretation.

Model · syllabus-based

39. Derive the vacuum plane-wave energy/intensity relations and use momentum transport to explain radiation pressure for an absorber and reflector.

Model · syllabus-based

40. Derive the wave equation and field ratios in a source-free homogeneous linear lossless nondispersive medium, and identify the limits of the model.

Model · syllabus-based

Coverage audit

40 numbered questions; 40 primary targets. The list below makes the intended distinction explicit.

Q1 · **T1** Direction from phase

Q2 · **T1** Meaning of wave equation

Q3 · **T2** Frequency wavelength wave number

Q4 · **T2** Phase difference

Q5 · T3 Polarization definition

Q6 · T3 Circular polarization conditions

Q7 · T3 Linear vector composition

Q8 · T3 Elliptical special case

Q9 · T4 Vacuum speed prediction

Q10 · T4 Source-free assumption

Q11 · T5 Field handedness

Q12 · T5 Magnetic amplitude

Q13 · T5 Transversality

Q14 · T6 Poynting vector units

Q15 · T6 Work density sign

Q16 · T7 Equal energy shares

Q17 · T7 Peak and rms distinction

Q18 · T8 Absorption pressure

Q19 · T8 Reflection factor

Q20 · T9 Medium speed and index

Q21 · T1 Arbitrary travelling waveform verification

Q22 · T2 Recovering a waveform from specifications

Q23 · T3 Eliminating phase for polarization ellipse

Q24 · T3 Polarization rotation with explicit viewpoint

Q25 · T4 Electric wave equation derivation

Q26 · T4 Magnetic wave equation derivation

Q27 · T5 Deriving the amplitude link

Q28 · T5 Testing an inconsistent field pair

Q29 · T6 Energy balance in a region

Q30 · T7 Intensity from electric peak

Q31 · T7 Field amplitude from beam power

Q32 · T8 Radiation force calculation

Q33 · T8 Partial reflection pressure

Q34 · T9 Impedance and field ratio in medium

Q35 · T9 Frequency across an interface

Q36 · T9 Energy transport speed in linear medium

Q37 · T4 Full vacuum plane-wave construction

Q38 · T6 Derivation and interpretation of Poynting theorem

Q39 · T8 Wave energy momentum and pressure synthesis

Q40 · T9 Propagation in homogeneous linear matter

Past-paper accounting

No question from the supplied local past papers maps directly to this module. All 40 are model questions, guided by the syllabus; no invented PYQ provenance.